



Pool boiling critical heat flux studies of accident tolerant fuel cladding materials

Hwasung Yeom^{a,*}, Emilio Gutierrez^a, HangJin Jo^b, Yimin Zhou^a, Kaya Mondry^a, Kumar Sridharan^{a,c}, Michael Corradini^a

^a Department of Engineering Physics, University of Wisconsin-Madison, WI 53706, USA

^b Department of Mechanical Engineering & Division of Advanced Nuclear Engineering, POSTECH, Pohang, South Korea

^c Department of Material Science and Engineering, University of Wisconsin-Madison, WI 53706, USA

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ABSTRACT

Pool boiling experiments have been performed at atmospheric pressure on potential accident tolerant fuel (ATF) cladding materials (Zirlo®, Cr coated Zirlo®, FeCrAl coated Zirlo®, and monolithic SiC) to evaluate the Critical Heat Flux (CHF) points. Corrosion resistant coatings of Cr and FeCrAl alloy were deposited on flat Zirlo® samples using cold spray technology. The as-prepared samples after uniform surface polishing were subjected to autoclave tests at 360 °C water and 18.6 MPa for 360 h to simulate prototypic corrosion of the ATF cladding materials in LWR normal operation. Surface characteristics potentially influencing CHF such as surface morphology, roughness, static contact angle, and surface chemistry were characterized using a suite of characterization methods including scanning electron microscopy, 3D optical profilometry, optical contact angle measurements, and x-ray photoelectron spectroscopy (XPS). Thermo-physical properties of the samples such as density, thermal conductivity, and heat capacity were measured by differential scanning calorimetry and laser flash thermal diffusivity measurement. FeCrAl coatings showed CHF values comparable to bare Zirlo® samples, while slightly lower CHF values were observed for Cr coated samples and bulk SiC flats (produced by chemical vapor deposition, CVD). As-deposited Cr coatings showed 67% higher CHF than the as-polished Cr coatings due to its higher surface roughness level. The autoclave testing for the prepared samples generally increased the CHF except for the surface polished Cr coatings. Similar CHF values of bare Zirlo®, FeCrAl coating, and CVD SiC samples were observed after autoclave test but there were negligible change CHF of Cr coatings. XPS studies indicated that the formation or deposition of a few monolayers of hydrophobic carbonaceous species on surface can affect CHF values. The trends in CHF data are discussed in terms of evolution of the surface characteristics of the ATF materials.

1. Introduction

Zirconium alloys have been widely used as fuel cladding materials for light water reactors (LWRs) on account of their neutron transparency, good mechanical strength, and corrosion resistance under normal LWR operating conditions (Zinkle et al., 2014). However, the Fukushima-Daiichi accident in 2011 revealed the limitations of Zr-alloys in accident scenarios where high temperature steam oxidation led to generation of heat and hydrogen, and destruction of the reactor core. In response to this accident, the U.S. Department of Energy (DOE) in conjunction with universities, national laboratories, and nuclear industries embarked the Accident Tolerant Fuel (ATF) program in 2012

(Carmack et al., 2013). As the near-term approach, Zr-alloys with thin protective coatings are being developed to provide additional safety and operational margin by virtue of their superior corrosion/oxidation resistance. The designs being considered for implementation by nuclear fuel vendors include a thin Cr or ceramic coatings on the conventional Zr-alloy fuel cladding using cold spray technology (CST) and physical vapor deposition (PVD) (Geelhood and Luscher, 2019). The long-term ATF designs, however, involve the complete replacement of Zr-alloy cladding with new materials such as FeCrAl alloys and SiC-SiC_f composites.

Development of ATF designs to date have focused on manufacturing processes to improve oxidation resistance, mechanical strength, and

* Corresponding author at: Engineering Physics Department, 1500 Engineering Drive Rm. 911, University of Wisconsin, Madison, WI 53706, USA.

E-mail address: hyeom@wisc.edu (H. Yeom).

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structural integrity in accident conditions. There is still uncertainty in regards to thermal-hydraulic performance of the ATF cladding concepts under prototypical LWR operating conditions (Su et al., 2020). For example, thermal limits such as critical heat flux (CHF) or departure of nucleate boiling (DNB) of ATF designs have not been thoroughly investigated, which are essential data for future licensing activity (Geelhood and Luscher, 2019). At the CHF point in LWR operation, fuel temperature abruptly increases due to limited heat transfer capability, resulting in degradation of mechanical integrity of reactor core. Therefore, accurate assessment of CHF values of ATF cladding concepts in various operational conditions must be performed to ensure safety margin for LWRs.

There have been a number of studies to establish experimental data of boiling heat transfer performance of ATF cladding candidate materials in various boiling conditions, sample geometries, and surface conditions. Son et al. (2015) fabricated a thin Cr coating and porous Cr_2O_3 coating on Ni-Cr wire of 0.5 mm diameter and 55 mm length using DC sputtering (a PVD technique) and nanofluid boiling deposition method, respectively. The pool boiling experiment showed that CHF value of the porous Cr_2O_3 coated samples was about two times higher than uncoated wires while Cr coated samples exhibited slight decrease in CHF. The CHF changes were explained by wetting state of the coated surfaces. Kam et al. (2015) reported pool boiling CHF values for SiC and Cr coated stainless steel, uncoated stainless steel, and Zircaloy-4 (Zirc-4) samples. SiC was deposited by sputtering technique and Cr was electroplated on the stainless steel substrates with dimension of 50 mm $\times$ 15 mm $\times$ 1.2 mm. The results of this study indicated that the thin SiC coating (1 μm thickness) increased CHF values by more than 56% but the Cr coated sample exhibited 29% lower CHF values compared to Zirc-4 samples. These observations were supported by wettability studies of surfaces as quantified by static water contact angle measurements. Boiling performance of PVD Cr, CrAl, and FeCrAl coated stainless steel substrates with modulation of micro-/nano-scaled surface topography have been extensively studied (Seo et al., 2016a, 2016b; Son et al., 2020, 2019, 2017). All coated surfaces generated by the sputtering technique inherently consisted of fine nano-scale surface topographical structure. The surface features of the coatings showing both nano-scaled structure (controlled by sputter deposition parameters) and micro-scaled structure (modulated by surface roughness of substrates) provided super-hydrophilic characteristics, resulting in enhancement of the CHF value up to 80% on flat samples (45 mm $\times$ 10 mm $\times$ 2 mm) and 30% on vertical tube sections (D9.5 mm $\times$ L80 mm $\times$ 0.88 mm), respectively, compared to uncoated reference, in a pool boiling condition. Ali et al. (2018) investigated corrosion effects of Fe-13Cr-4Al alloy on surface characteristics and pool boiling CHF. The FeCrAl samples oxidized under simulated LWR conditions exhibited an increase in surface roughness and wettability compared to as-machined samples. The measured CHF value of oxidized FeCrAl alloy samples (10 mm $\times$ 10 mm $\times$ 0.5 mm) under PWR and BWR condition for one year were 35% and 20% higher, respectively, than Zirc-4 samples. In addition, the effects of ion irradiation on surface wettability and CHF of Cr-coated Zirc-4 in pool boiling condition were also studied by Ali et al. (2020). Cr was deposited on flat samples by cold spray technology followed by cold rolling to control coating thickness and surface roughness. Although high energy ion irradiation on the coated samples altered its surface energy slightly, CHF values did not significantly change compared to pre-irradiated surfaces. Finally, Jo et al. (2019) conducted pool boiling experiment for Cr coated, FeCrAl coated, uncoated Zirlo®, and stainless steel flats (20 mm $\times$ 20 mm $\times$ 0.8 mm) in pool boiling condition. The coatings (40–80 μm thickness) were produced by cold spray technology, where the surfaces were polished to match the fuel cladding surface roughness. The study concluded that surface parameters and CHF values were not changed significantly due to the coatings because of negligible differences in surface roughness and wettability after the surface finishing process, although thermo-physical properties of the near-surface regions of the material could also influence the CHF value.

In summary, the majority of experimental investigations to evaluate thermal limit of ATF candidate materials such as Cr coatings, FeCrAl bulk/coatings, and SiC coatings were performed with horizontal flat geometry of clean samples in pool boiling conditions at atmospheric pressure. The results showed that surface characteristics including wettability (measured by water droplet contact angle) and surface roughness/morphology are important parameters affecting CHF values. Cumulative surface effects related to capillary wicking and active nucleation sites would impact boiling heat transfer coefficient and CHF (Kim et al., 2016, 2015). Kandlikar (2001) developed a theoretical model to predict CHF values as a function of receding contact angle based on a force balance between evaporation momentum inertia and surface tension forces. However, more rigorous experiments are required to arrive at conclusive data on CHF for ATF candidate materials, since there are some discrepancies in CHF values of even the same material samples. This is likely due to different surface conditions related to surface finishing and sample manufacturing processes. Moreover, it is important to incorporate evolution of surface composition and topography of ATF materials under reactor operation into estimation of the CHF point. Pre-oxidizing ATF materials separately in prototypical reactor conditions prior to boiling tests would allow for a direct comparison of the data with fresh unoxidized fuel cladding condition. Such a separate effects test would provide fundamental insights into the correlation between surface evolution by water corrosion and the CHF phenomenon.

The present study investigates water corrosion effects on evolution of surface characteristics and pool boiling CHF values of ATF candidate materials. Cold spray technology was used to fabricate Cr and FeCrAl coatings on Zirlo® flats. The coated Zirlo®, uncoated Zirlo® samples, and CVD (chemical vapor deposition) SiC samples were subjected to 360 h water autoclave test to simulate surface conditions of fuel cladding in LWR reactor conditions. Surface characteristics of the samples including morphology, wettability, roughness, chemistry, and thermo-physical properties were quantified by a suite of near-surface materials characterization methods. Finally, pool boiling experiments for the fresh samples and autoclave tested samples were conducted at atmospheric pressure to measure CHF points which were then compared with published data.

2. Experimental procedures

2.1. Material procurement, cold spray process, and water corrosion test

Bulk Zirlo® sheet with thickness of 0.8 mm used as the substrate for Cr and FeCrAl cold spray coatings were supplied by Westinghouse Electric Company (PA, USA). Monolithic CVD β -SiC coupons with dimension of 20 mm $\times$ 20 mm $\times$ 1 mm were purchased from PremaTech Advanced Ceramics (MA, USA) and had a purity level of greater than 99.9995%. The CVD SiC samples were used as surrogate for SiC-SiC_f composite designs. It is noted that SiC-SiC_f composite cladding designs are engineered to have an outer CVD SiC layer for protection from direct exposure of the SiC fiber matrix to the water coolant (Deck et al., 2020).

A commercial Cold Gas Technologies (CGT) Kinetiks 4000/34 unit system at the University of Wisconsin, Madison was utilized for deposition of dense Cr and Fe-20Cr-5Al coatings on the Zirlo® substrate with previously optimized parameters (Dabney et al., 2019; Maier et al., 2019). Only the front side of the strips were coated, sectioned approximately 20 mm square coupons, and then polished with 600 grit SiC paper to reduce the surface roughness. The resultant Cr coatings and FeCrAl coatings had a thickness range of 30 to 70 μm . The same surface finish was applied to uncoated Zirlo® coupons which were served as the baseline samples in water corrosion and pool boiling tests. The SiC coupons were polished with a 30 μm diamond-embedded nickel polishing disc to remove naturally formed oxide layer. Surfaces of all flat samples were thoroughly cleaned by sonicating in sequentially in acetone and ethanol baths for 5 min each, before any tests. Samples after

autoclave corrosion tests were dried in ambient air for 24 h to remove any retained water on the surface prior to surface examinations and pool boiling tests.

Isothermal corrosion tests in high temperature water, simulating corrosion of materials in LWR coolant environment, were performed for the coated and the bulk coupons. The test was performed using a 4650 series autoclave from Parr Instrument Company (IL, USA) with a total test volume of 1 L and test section made of nickel-based alloy 625, in accordance with the ASTM G2 standard (ASTM Int, 2019). The maximum sample volume in the autoclave system is within the limit of ASTM standard. The samples were tested in saturated water at 360 °C and 18.6 MPa for 15 days (360 h). Static deionized pure water (18.2 MΩ-cm) with 5 ppm dissolved oxygen was used for these tests. Table 1 summarizes sample conditions for boiling tests used in this study.

2.2. Pool boiling facility and test procedure

A pool boiling test facility, consisting of a water chamber, an immersion heater, a Cu heater block with thermocouples, and a sample holder, was designed to perform CHF experiment for flat samples (20 mm × 20 mm) of various thickness. DI water (18.2 MΩ-cm) filled in a rectangular pool (140 mm × 140 mm × 140 mm) made of aluminum alloy was maintained at saturated condition during the experiment by an immersion heater. Fig. 1 shows a schematic diagram of the assembly of the test section. A flat sample of the materials was fixtured at their edges with high temperature epoxy (3 M Scotch-Weld Epoxy Adhesive DP460) to the polyether ether ketone (PEEK) sample holder, installed at the bottom of the water chamber using a water-tight silicone gasket. With the top surface of the sample exposed to saturated water, four cartridge heaters (500 W each) inserted inside the Cu heater block generated heat, which was then conducted up through the block and through a high temperature thermal paste (2.3 W/m·K, Omegatherm®) on the underside of the test sample. Constant contact between the heating sample and the Cu heater block during boiling tests was provided by a mechanical supporting system (i.e., supporting jack and rubber buffer sheet) underneath the Cu heating block. Additional information of the experimental facility is provided in our previous paper (Jo et al., 2019).

To begin the boiling experiment, any dissolved gases in the working fluid was removed by maintaining saturated temperature for 1 h. After the degassing step, voltage supplied to the cartridge heaters was increased in a step-wise manner to avoid abrupt transition in boiling while three 0.5 mm OD K-type thermocouples (OMEGA engineering) installed at 5 mm intervals monitored temperature of the heater block. The applied heat flux was estimated using 1D Fourier's conduction equation ($q'' = k\Delta T/\Delta x_{TC}$). We assumed the heat loss in the Cu heater block to environment to be negligible due to the high thermal conductivity of the copper. Moreover, the linear temperature relationship between the thermocouples was confirmed with a numerical simulation for the steady-state heat conduction in the actual heater geometry using ANSYS software (2019 R1). This heating process continued until a rapid

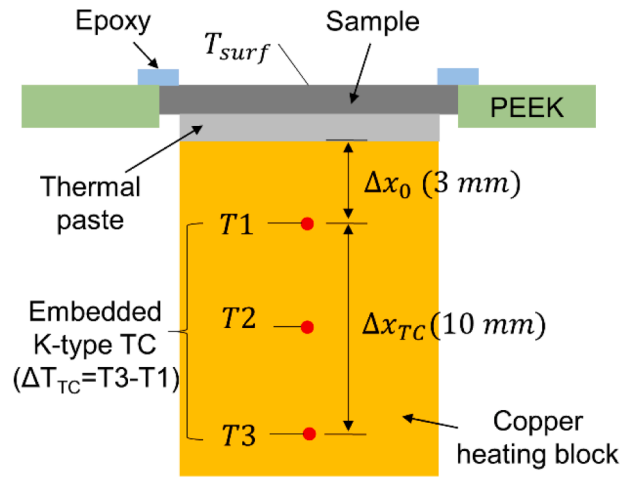


Fig. 1. Schematic diagram of the test section of the pool boiling CHF test facility used in this study (not to scale).

temperature increase in the thermocouples associated with a sudden change in heat transfer coefficient and visual changes in bubble dynamics were observed. Surface temperature (T_{surf}) of samples was estimated by 1D steady-state conduction model with a known thermal conductivity and thickness of the test section components (sample, thermal paste, Cu block) and calculated heat flux. The thickness of the thermal paste was assumed to be constant (0.2 mm) since the paste was rapidly cured during boiling test and any dimensional change of the Cu heating block due to the thermal expansion was compensated by the elastic silicone buffer layer in the supporting system. Finally, the maximum uncertainty of heat flux measurement was estimated to be less than 1% as described in our earlier paper (Jo et al., 2019).

2.3. Materials surface characterization

Surface characteristics, including surface chemistry and topography of the as-prepared samples and those subjected to autoclave and pool boiling tests, were investigated by a series of characterization methods. Sample roughness was quantified using a 3D optical profilometer (Zygo New View 6300). Two types of surface profile parameters, the arithmetic average height (R_a) and the maximum peak-to-valley height (PV), were determined from measurements at three different locations on each sample with surface area of 0.70 mm × 0.53 mm. Surface morphology of the samples was characterized by secondary electron imaging in a Zeiss LEO scanning electron microscope (SEM). A Zeiss Auriga FIB (focused ion beam) was used to mill cross-sectional trenches to evaluate oxide layer thickness on sample surfaces after the autoclave tests. Static contact angle of samples was measured using Dataphysics OCA 15 Optical Contact Angle measuring system in accordance with ASTM D7334-08 standard (ASTM Int, 2008). Apparent static contact angle was reported in this study although dynamic receding contact angle is considered to be a more representative parameter to elucidate the relationship between contact angle and CHF (Kandlikar, 2001). For the static contact angle measurements, a DI water droplet (5 μl) was dispensed onto the surface while recording the contact angle of the water droplet for three seconds. The short measurement time was expected to minimize evaporation or diffusion effects. The contact angle of a surface was determined by the average of three measurements of individual droplets. X-ray Photoelectron Spectroscopy (XPS) was used to identify evolution of very near-surface chemistry of the samples during the autoclave and the pool boiling tests. The XPS investigation was performed in the Thermo k-alpha x-ray photoelectron spectrometer equipped with monochromatic Al k-alpha x-ray source with a 400-μm spot size. A few selected samples from the pool boiling tests and the autoclave corrosion tests were also subjected to incremental argon ion

Table 1

Summary of sample conditions subjected to the boiling test used in this study where “As”, “Po”, “Au” stands for the samples with as-deposited, surface polished, and autoclave tested, respectively.

Sample ID	Heater material	Surface finish	Test
Zir-Po	Zirlo®	600 grit SiC	–
Zir-Po-Au	Zirlo®	600 grit SiC	Autoclave test
Cr-As	Cold spray Cr coated Zirlo®	As-deposited	–
Cr-Po	Cold spray Cr coated Zirlo®	600 grit SiC	–
Cr-Po-Au	Cold spray Cr coated Zirlo®	600 grit SiC	Autoclave test
FCA-Po	Cold spray FeCrAl coated Zirlo®	600 grit SiC	–
FCA-Po-Au	Cold spray FeCrAl coated Zirlo®	600 grit SiC	Autoclave test
SiC-Po	CVD SiC	30 μm diamond	–
SiC-Po-Au	CVD SiC	30 μm diamond	Autoclave test

beam sputtering (ion energy of 2 keV) and XPS measurements to obtain depth-resolved compositional in the near-surface regions of the samples. To provide a better understanding of chemical binding energy of element of interest (and hence its chemical state), identification of peaks of binding energy for the samples was conducted via commercial software (Origin 2018b).

2.4. Thermo-physical properties

It is known that thermal activity (S) of heaters, a product of sample thickness (δ) and thermal effusivity ($\sqrt{\rho C_p k}$), correlates to CHF. CHF increases as the thermal activity increases until a certain asymptotic value (i.e., hydrodynamic instability limit) (Arik and Bar-Cohen, 2003; Golobič and Bergles, 1997). Local temperature of a heater underneath vapor blanket near CHF is controlled by radial thermal diffusion, as a function of the thermal activity. It was postulated that local dry patches on thin heaters with poor thermal properties can reach the Leidenfrost temperature so that the returning liquid is unable to quench the local surface effectively. On the other hands, a heater with high thermal activity successfully absorbs and conducts the heat away from the local dry spots in boiling process, resulting in rewetting the dry spots and sustaining nucleate boiling regime. Therefore, hydrodynamic instability determines CHF on such heaters where unstable liquid and vapor interface limits the liquid flow to the heater surface. Thermo-physical properties such as thermal conductivity (k), heat capacity (C_p), and density (ρ) of selected flat samples (Zirlo® flat with 100 μm Cr coating, uncoated Zirlo®, CVD SiC) at two temperatures (25 $^{\circ}\text{C}$ and 200 $^{\circ}\text{C}$) were measured in CMC Laboratories (AZ, USA). Density of the flat samples was determined via Archimedes principle using a silicon standard reference. Heat capacity of the samples was directly measured using differential scanning calorimetry (DSC) in accordance with ASTM standard E1269 (ASTM Int, 2001). Difference of heat flow between as-prepared samples and a reference sapphire with known heat capacity in controlled nitrogen atmosphere was monitored to calculate the heat capacity. Thermal conductivity of the samples was evaluated from the measured density, heat capacity, and thermal diffusivity (α) using laser flash thermal diffusivity measurements. In this method, a laser pulse heats the front side of sample surface while temperature of the back-surface is recorded to determine the thermal diffusivity based on one-dimensional heat flow model. Finally, thermal conductivity was calculated using measured parameters ($k = \rho \cdot C_p \cdot \alpha$). The test procedure was reliable since measured values of reference materials (Al, Cu, C) were typically within 2% of published literature values and NIST traceable standards.

3. Results

3.1. Surface characteristics

3.1.1. Surface morphology and cross-sectional examination

Fig. 2 shows plan-view SEM micrographs of uncoated Zirlo®, polished and unpolished Cr coated Zirlo®, and CVD SiC samples before and after the autoclave corrosion test. As-polished samples (Zir-Po, Cr-Po, FCA-Po, SiC-Po) revealed smooth surfaces with fine scratches on the surfaces from the polishing process. The surfaces of Zirlo® samples appeared rougher than Cr coated samples due to the lower hardness although the same polishing procedure was used. Pre-existing porosity in CVD SiC matrix left a number of surface cavities on the surface after the surface polishing process. As-deposited cold spray coatings inherently revealed rough surfaces, as shown in Fig. 2d, since the coating formation relies on stochastic impact of powder particles on a substrate followed by the plastic deformation. The autoclave corrosion test changed surface morphology of uncoated Zirlo® samples so that surface cracks, bulges, and spallation of oxide layer with debris were identified on local areas (Fig. 2e), whereas no significant morphological change of Cr coated samples was observed. As shown in Fig. 2h, high magnification SEM images for the FeCrAl coated samples showed nano-sized particles/whiskers on the surfaces, which was also observed in a past work (Terrani et al., 2016).

Cross-sectional micrographs (Fig. 3) of the autoclave tested samples showed that an oxide layer (1 to 1.3 μm thickness) formed on the uncoated Zirlo® samples while a very thin oxide layer (20 to 30 nm thickness) formed on the Cr coatings. This result indicates that the surface coatings enhanced the corrosion resistance to the underlying Zirlo® material in the prototypical LWR operation condition.

3.1.2. Surface roughness

Surface roughness values (R_a and PV) for three randomly selected areas on each sample were measured and then averaged with multiple samples (more than three samples). The measurement was also performed for samples after the boiling test to evaluate the evolution of surface roughness induced by the vigorous boiling phenomenon. All measured roughness values are presented in Table 2. The as-deposited Cr coatings exhibited very high roughness; the R_a value was eighty times higher than that of the as-polished Cr coatings. The surface polish process normalized surface roughness so that the polished surfaces exhibited the R_a range between 0.07 μm and 0.14 μm . It is noted that uncoated Zirlo® samples were more prone to have polishing scratches due to lower hardness than the coatings. In general, the autoclave corrosion test increased surface roughness of all samples to some extent. The large increase in R_a and PV of uncoated Zirlo® samples after the corrosion test corresponds to the surface irregularities such as cracking,

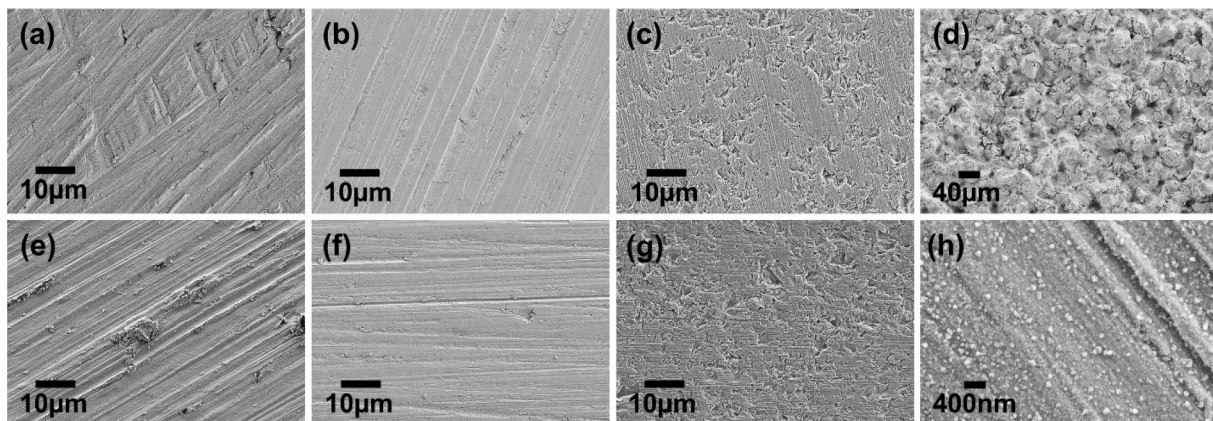


Fig. 2. Plan-view surface SEM micrograph for as-prepared samples: (a) Zir-Po, (b) Cr-Po, (c) SiC-Po, and (d) Cr-As, and autoclave tested samples: (e) Zir-Po-Au, (f) Cr-Po-Au, and (g) SiC-Po-Au. and (h) higher magnification of FCA-Po-Au surface showing nano-scaled oxide particles.

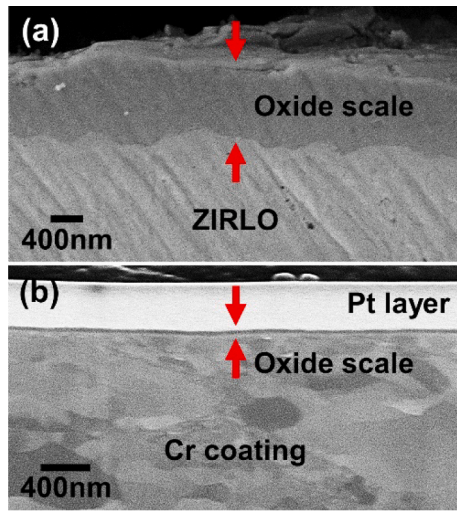


Fig. 3. Cross-sectional SEM image of (a) autoclave tested Zirlo® sample (Zir-Po-Au) and (b) autoclave tested Cr coating (Cr-Po-Au) using FIB cross-sectional milling technique. Oxide scale is indicated by the red arrows. A thin Pt layer was deposited on the Cr coating to protect the oxide layer from any damage during the milling process. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Summary of average roughness values for the prepared samples where “BT” stands for the samples after the boiling test. S.D represents standard deviation.

Test sample	Surface roughness (μm)	
	R_a (S.D.)	PV (S.D.)
Zir-Po	0.12 (0.02)	3.13 (0.04)
Zir-Po-Au	0.21 (0.06)	7.56 (2.83)
Zir-Po-Au-BT	0.21 (0.04)	7.84 (3.77)
Cr-As	5.72 (0.40)	80.93 (9.87)
Cr-Po	0.07 (0.01)	3.14 (0.52)
Cr-Po-Au	0.17 (0.02)	4.17 (1.17)
Cr-Po-Au-BT	0.17 (0.03)	4.77 (1.81)
FCA-Po	0.07 (0.01)	2.94 (0.34)
FCA-Po-Au	0.15 (0.05)	7.86 (7.55)
FCA-Po-Au-BT	0.14 (0.04)	8.58 (8.25)
SiC-Po	0.14 (0.01)	5.02 (0.49)
SiC-Po-Au	0.15 (0.02)	5.36 (0.66)
SiC-Po-Au-BT	0.15 (0.02)	4.89 (1.06)

spallation, and bulge of the oxide scale due to the compressive stresses. The roughness parameters of the polished Cr coatings were also increased slightly but the values were still smaller than other uncoated samples. It is noted that FeCrAl coatings revealed the high PV value after the autoclave test. Based on SEM images, corrosion of inter-particle boundary in the FeCrAl coatings led to formation of localized micron-sized craters. The surface defects were responsible for the increased PV values. No clear change in roughness was evident for the CVD SiC samples during the 360 h autoclave exposure. In addition, the effect of boiling experiment on the surface roughness parameters was negligible.

3.1.3. Static water contact angle

Static contact angles for as-prepared samples (θ_{AS}) and for samples after boiling test (θ_{BT}) are summarized in Table 3. The apparent static contact angle will be used to establish a qualitative relationship between surface wettability and CHF value with the assumption that both dynamic contact angle and static contact angle increase or decrease together (Ali et al., 2018). For each sample, three measurements were collected randomly on the entire area of the surface, averaged from multiple samples to obtain a statistically significant representation.

For the as-polished metallic surfaces, the contact angle was in the

Table 3

Summary of average static contact angle for the as-prepared samples before and after the boiling test.

Test sample	Static contact angle (degree)	
	As-prepared, θ_{AS} (S.D.)	After boiling test, θ_{BT} (S.D.)
Zir-Po	72 (13)	79 (2)
Zir-Po-Au	52 (10)	77 (3)
Cr-As	107 (9)	n/a
Cr-Po	77 (6)	77 (2)
Cr-Po-Au	93 (5)	77 (7)
FCA-Po	86 (5)	88 (2)
FCA-Po-Au	79 (9)	74 (4)
SiC-Po	33 (6)	105 (2)
SiC-Po-Au	77 (15)	83 (4)

range of 70° to 90°. The autoclave test altered the wettability of the surfaces. The contact angles decreased for Zirlo® and FeCrAl coatings, but increased for the Cr coatings and CVD SiC, after the autoclave corrosion test. The effect of autoclave corrosion test on wettability of uncoated Zirlo and FeCrAl samples is in good agreement with published data for Zircaloy-4 and FeCrAl alloys subjected to long-term LWR corrosion (Ali et al., 2018). It is believed that development of an oxide scale on the samples is responsible for the increased wettability, which is supported by contact angle data for oxidized FeCrAl alloys and Zircaloy-4 (Lee et al., 2019a, 2019b, 2019c). However, to the best of the authors' knowledge, the increase in contact angles for Cr coatings and CVD SiC after autoclave corrosion test has not been reported yet. The as-deposited Cr coatings showed very high contact angle compared to the polished Cr coating, an effect that is attributed to the large surface roughness differences. On the basis of modified Wenzel's wetting model proposed by Yu et al. (2019), roughened surface magnifies nonwetting characteristics of solid surfaces with an intrinsic contact angle range of 43° to 90°. Therefore, it is reasonable to say the increased surface roughness of the Cr coatings could increase the contact angle.

The contact angles of the samples increased marginally after a boiling test except for autoclave corrosion tested Zirlo® and Cr coatings, and polished CVD SiC. It is speculated that the contact angle changes are related to aging effect on the surface chemistry of the samples or a certain surface interaction with water during boiling. The aging effect on the wetting behavior of PVD Cr coatings after air/DI water exposure has been reported (Umretiya et al., 2020a). The as-polished CVD SiC revealed an abrupt increase in contact angles both after the boiling test (increase by 72°) and the autoclave corrosion test (increase by 44°), respectively. It is hypothesized that the very near-surface chemistry of CVD SiC was altered during the tests, which has been investigated by XPS surface compositional analysis reported in the following section.

3.1.4. XPS compositional analysis

XPS studies were performed on the autoclave tested samples. After no unexpected surface species were identified in a wide energy range survey spectrum, narrow energy region core level spectra of each type of samples were collected. Compositional depth profile was obtained using 2 keV Ar^+ ion beam sputtering for 30 s between binding energy acquisitions. The ion beam sputtering was performed with fifty intervals (total sputter time is 3600 s). Although the resultant depth scales were too shallow to be quantified by the optical profilometer, it is known that the relative sputter rate depends on material composing the surface (e.g., sputter rate under a certain ion beam condition: $\text{C} > \text{SiC} > \text{SiO}_2 > \text{Fe}_3\text{O}_4 > \text{Cr}_2\text{O}_3$) (Baer et al., 2010; McIntyre and Zetaruk, 1976; Sathiyamoorthy and Weisweiler, 1982). The compositional depth profiles of the autoclave tested samples are shown in Fig. 4. The autoclave test resulted in oxidation of the metallic samples to different extents. The high oxygen concentration in the Zirlo surface was clearly evident during the entire duration of sputtering but sudden drop of oxygen content on both Cr and FeCrAl surfaces were observed after sputtering for 2,000 s. This is in good agreement with the thickness of the oxide layers shown in cross-

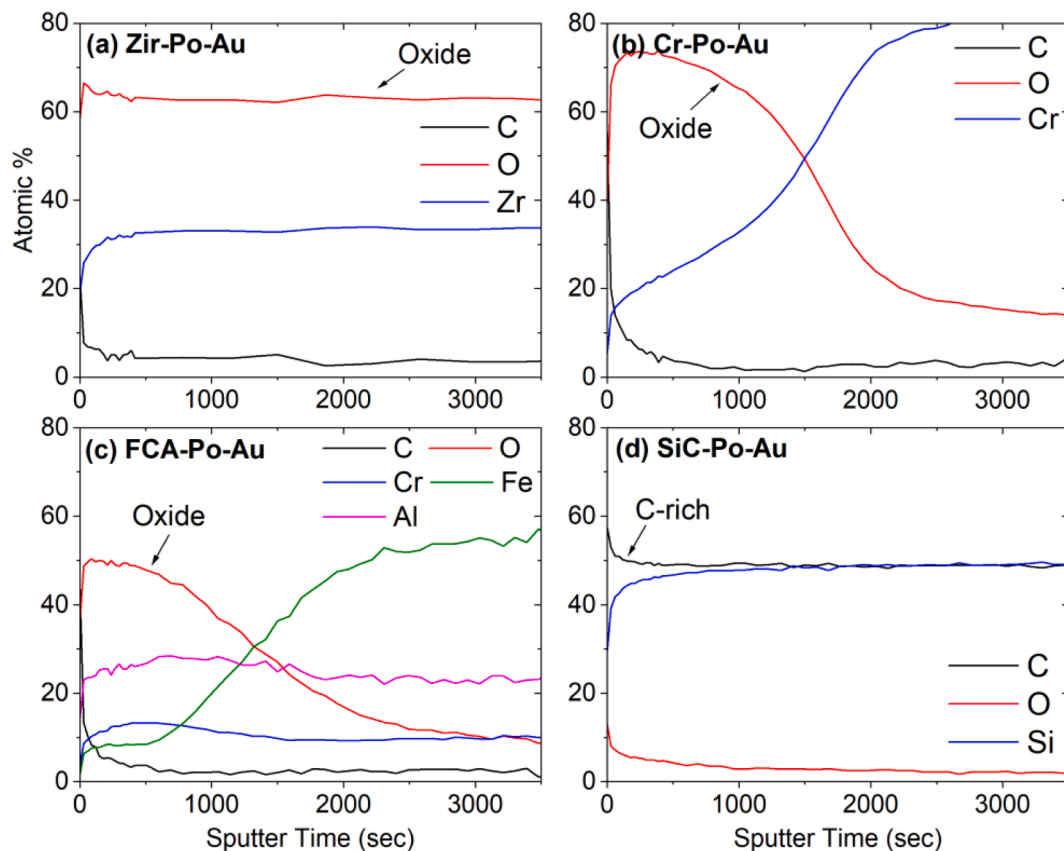


Fig. 4. Composition vs. depth (sputter time) profile on the surfaces of (a) uncoated Zirlo, (b) Cr coated Zirlo, (c) FeCrAl coated Zirlo, and (d) CVD SiC sample after the autoclave corrosion test. Ar^+ ion beam sputtering energy for analysis was 2 keV. Elements of interest were chosen depending on the composition of each surface.

sectional SEM examinations (Fig. 3). In addition, it is speculated that thickness of the Cr_2O_3 layer on the Cr coating was lower than the multi-components oxide layer including Fe_3O_4 formed on the FeCrAl surface based on the different sputtering rate of the oxides. Interestingly, no indication of Si-oxide layer formation on the CVD SiC was observed after the autoclave test, but rather a very thin C-rich layer was identified. This result is consistent with XPS data reported on corrosion tested SiC specimens tested in LWR conditions (Terrani et al., 2015). The silica formed on SiC surfaces under hydrothermal conditions readily dissolves in water (Yeom et al., 2018). It is not clear whether the thin C-rich layer was a corrosion product in chemical reactions between SiC and water during the autoclave test or physical deposition of carbonaceous compounds during sample handling/storage.

The accumulation of carbon contaminants on sample surfaces was evident in XPS C1s spectra as shown in Fig. 5. Adventitious carbon forms such as hydrocarbon polymers (C-H) and graphitic system (C-C), corresponding to binding energy range between 284.4 eV and 285.3 eV (Barr and Seal, 1995; Greczynski and Hultman, 2017), were identified in all autoclave tested sample surfaces. It is noted that the intensity of the adventitious carbon species varied depending on the surface chemistry. The Cr coating showed the highest count rate of the adventitious carbon signal but the Zirlo® surface revealed the lowest count rate. It is important to mention that this trend is consistent with the contact angle data of the autoclave tested samples (Table 3). Again, Umretiya et al. (Umretiya et al., 2020b) reported hydrophobic wetting characteristic of Cr coatings was resulted from the deposition of the organic contaminants from water and air environments.

XPS data for CVD SiC surfaces were examined to elucidate the reason for increase in contact angle of the as-polished SiC after a boiling test. Fig. 6 shows the core level spectra for Si2p, O1s, and C1s on the surface of CVD SiC sample after the boiling test. The distinct binding energy

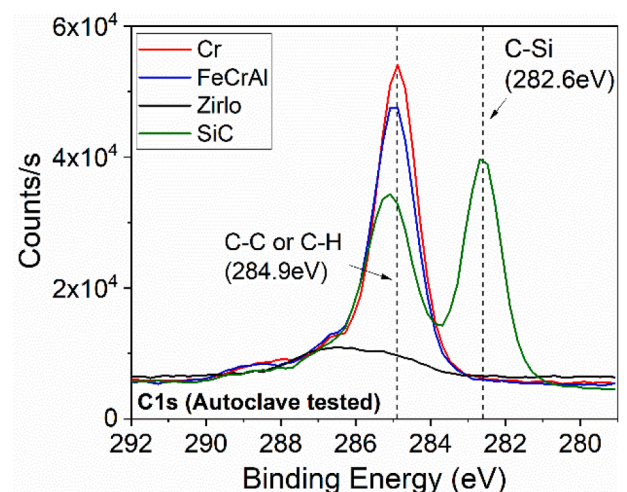


Fig. 5. C1s XPS spectra acquired from uncoated Zirlo, Cr coated Zirlo, FeCrAl coated Zirlo, and CVD SiC sample after the autoclave corrosion test. Ion sputtering on the surfaces was not performed.

peaks were labeled based on published data (Contarini et al., 1991; Iwanowski et al., 1999). The surface chemistry of the as-polished CVD SiC surface consisted of SiC, native Si-suboxide (SiO_x), and a small amount of carbon contaminant. After pool boiling test, the chemical composition of the surface changed with the intensity of carbon peaks including hydrocarbon, graphite, and other organic compounds (hydroxyl and carbonyl carbon) increasing, while those of SiC and suboxide peaks decreased. This result indicates that organic compounds either formed or were deposited on the CVD SiC surface from environment (a

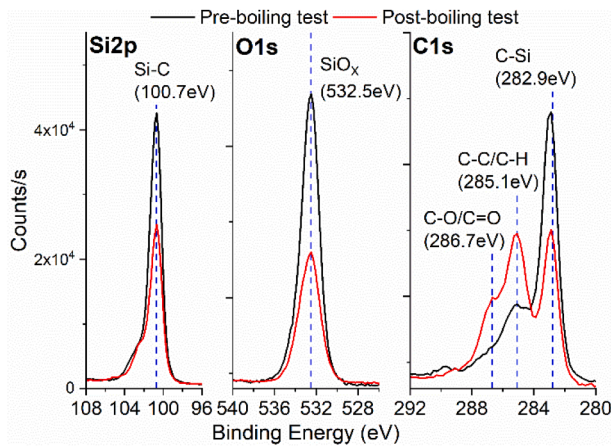


Fig. 6. XPS spectra (Si2p, O1s, and C1s) of the CVD SiC surface after a boiling test. The surface chemistry of SiC evolved during the pool boiling test.

pool boiling) or corrosion effects, which is attributed to the significant increase in the static contact angle (increase by 72°).

3.2. Thermo-physical properties

Table 4 summarizes thermo-physical properties including thermal conductivity, heat capacity, density, and thermal activity of uncoated Zirlo®, as-polished Cr coated Zirlo®, and as-polished CVD SiC coupon at two temperatures (24 °C and 200 °C). The measurements were conducted nine times for each sample and the experimental uncertainty was less than 0.1%. The measured values for the monolithic samples were in good agreement with the thermo-physical properties reported in literature (Krett et al., 1997; Luscher and Geelhood, 2014; Snead et al., 2007). It was noted that the thin Cr coating increased thermal conductivity and heat capacity only marginally since its thickness was approximately 11% of the total thickness of the coated sample. The CVD SiC samples showed more than ten times higher thermal conductivity and two times higher heat capacity than the metallic materials. Therefore, the thermal activity of CVD SiC is much higher than those of the bare and coated Zirlo® samples. Boiling tests were used evaluate significance of the sample material's thermal parameters on CHF, as reported in the next section.

3.3. Boiling curves and CHF

The pool boiling tests for the as-prepared materials including autoclave corrosion tested samples were performed to evaluate CHF points. At least three samples for each surface condition were tested to make ensure reproducibility of the measurements. The estimated boiling curves and a summary of averaged CHF values along with their standard deviations are presented in Fig. 7 and Table 5, respectively. The experimental uncertainty of CHF points for the as-polished Zirlo® and autoclave tested Zirlo® samples were 1.5% and 8% of the mean values, respectively.

Table 4

Summary of thermo-physical properties of bare Zirlo®, Cr coated Zirlo®, and CVD SiC coupon measured at room temperature and 200 °C. Literature values (Krett et al., 1997; Luscher and Geelhood, 2014; Snead et al., 2007) are also shown for comparison.

Material	Temperature (°C)	Thermal conductivity, k (W/m·K)		Heat capacity, C _p (J/kg·K)		Density, ρ (g/cm ³)		Thickness (mm)	Thermal activity (J/m·K)
		Meas.	Lit.	Meas.	Lit.	Meas.	Lit.		
Zirlo®	24	13.5	12.6	272.0	280.7	6.6	6.5	0.81	4.0
	200	15.5	15.0	313.8	301.6	6.6	–		4.6
Cr coated Zirlo®	24	15.0	n/a	313.8	n/a	6.7	n/a	0.91	5.1
	200	15.8	n/a	338.9	n/a	6.6	n/a		5.4
CVD SiC	24	323	354	732	670	3.2	3.2	1.06	29.1
	200	182	214	1029	944	3.2	–		25.9

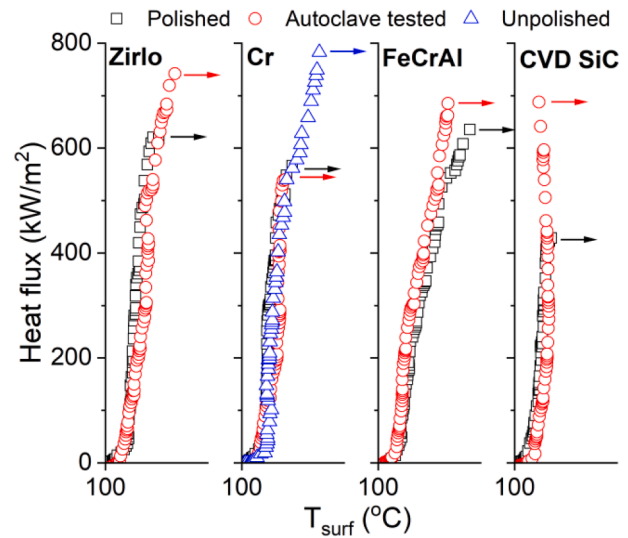


Fig. 7. Examples of boiling curves of Zirlo®, Cr coated Zirlo®, FeCrAl coated Zirlo®, and CVD SiC flat samples with different surface conditions (as-polished, un-polished, and autoclave tested). The surface temperature is not presented in the plots since it was not directly measured during the experiment.

Table 5

Mean and standard deviation of measured CHF values. The number of samples used for evaluating CHF point is also specified.

Sample ID	# of samples	CHF (kW/m ²)
Zir-Po	4	645 ± 10
Zir-Po-Au	11	814 ± 65
Cr-As	3	931 ± 94
Cr-Po	6	558 ± 32
Cr-Po-Au	5	550 ± 16
FCA-Po	3	630 ± 14
FCA-Po-Au	5	737 ± 58
SiC-Po	4	454 ± 50
SiC-Po-Au	3	667 ± 84

The measured CHF values of the as-polished metallic samples are consistent with our previous work (Jo et al., 2019). CHF points of FeCrAl coated Zirlo® samples were comparable to those of uncoated Zirlo® but Cr coatings exhibited 13% lower CHF values. However, as-deposited Cr coatings exhibited 67% higher CHF than the as-polished Cr coatings, due to their very high surface roughness. It is interesting to note that the lower CHF values were observed in as-polished CVD SiC samples compared to the metallic samples, which can be explained by its high contact angle after the boiling test. CHF enhancement was observed for all autoclave corrosion tested samples except the Cr coated samples. The CHF increase for Zirlo®, FeCrAl coated Zirlo®, CVD SiC samples after the autoclave test were approximately 26%, 17%, and 47%, respectively due to the surface corrosion. It is noted however that the actual CHF values of these surfaces were not significantly different. There was

negligible change in CHF values of Cr coated samples due to the autoclave corrosion test.

4. Discussion

4.1. Effect of surface wettability on CHF

Fig. 8 presents the measured CHF values from the present study, published CHF data for the ATF candidate materials, and predicted CHF values using the Kandlikar model (Kandlikar, 2001) with $\pm 20\%$ uncertainties, as a function of static contact angle. The measured CHF points were plotted using the post-boiling test static contact angle (θ_{BT}) as shown in Table 3. It appears that θ_{AS} does not reflect near-surface compositional changes in the material and associated wettability change induced by a boiling test. The static contact angles and pool boiling CHF values for zirconium-alloy samples in various surface conditions (e.g., polished, oxidized in air, corrosion in BWR or PWR conditions) were obtained from literature (Ahn et al., 2010; Ali et al., 2018; Kam et al., 2015; Lee et al., 2015; G. Lee et al., 2019). The data for as-prepared Cr coatings deposited by PVD technique, cold spray technology, and electroplating were taken from previous studies (Ali et al., 2020; Kam et al., 2015; Son et al., 2020). Boiling test data for stainless steels and FeCrAl alloys were also found in open literature (Ali et al., 2018; Seo et al., 2016a, 2016b; Son et al., 2020). Finally, one experimental data for a horizontal SiC heating surface was available in a paper investigating boiling performance of PVD SiC coated stainless steel flats (Kam et al., 2015).

It is believed that static contact angle is still useful as a qualitative measure of wettability of the heater surface although it is not the most representative quantity correlating to CHF phenomenon in pool boiling conditions. This is because a receding contact angle, a widely used contact angle parameter to elucidate CHF as a function of wettability, showed a trend similar to the static contact angle on copper surfaces with approximately 10° difference at room temperature (Kim et al., 2016). It is reasonable to perform a direct comparison between the measured data and the reported data in literature since all CHF data were acquired using horizontal flat samples in pool boiling water. Moreover, the surface roughness level of the materials in literature was in the R_a range of $0.1 \mu\text{m}$ to $0.6 \mu\text{m}$, encompassing the roughness range of the samples in the current study.

The CHF data measured in the present study and published data are in good agreement with predicted CHF values within $\pm 20\%$ uncertainty. This uncertainty level was reported in the experimental validation of the

theoretical model. This result indicates that surface wettability of the ATF candidate materials is a prominent factor dictating the CHF point. The moderate wettability of the heater surfaces (surfaces finished by a conventional polishing procedure) can be altered by environmental factors such as oxidation at high temperature and corrosion in LWR conditions, leading to change in CHF to different degrees depending on near-surface chemistry of material. For example, corrosion of FeCrAl alloys in high temperature water conditions led to decrease in static contact angle and increase in CHF while the Cr coatings did not show significant alteration of either static contact angle or CHF values after autoclave corrosion tests. A minimal change in contact angle of Cr coatings was also observed in three consecutive quenching experiments where pre-heated coated samples (600°C) were quenched in subcooled water (K. Lee et al., 2019). The authors concluded that limited oxidation of Cr coatings by virtue of its excellent corrosion resistance resulted in negligible change of surface wettability.

4.2. Effects of surface roughness on CHF

Roughness parameters (R_a , PV) of the as-prepared samples are plotted against the CHF values as shown in Fig. 9. The parameters exhibited a weak dependency with the measured CHF data, CHF increasing gradually with increasing surface roughness on the relatively smooth surfaces (i.e., $0.05 \mu\text{m}$ to $0.25 \mu\text{m}$ in R_a range). In addition, the very rough Cr coatings ($5.72 \mu\text{m}$ R_a) showed the highest CHF point in the experiment (not shown in the figure). The autoclave tested samples with higher roughness level generally showed higher CHF points compared to the as-polished samples. Roughness of heater surfaces is known to influence boiling heat transfer coefficient but the effect on CHF is still not clear (Kim et al., 2016). O'Hanley et al. (2013) performed pool boiling tests of surface engineered sapphire substrates with varying surface roughness by creating micro-posts using the photolithography technique. The authors concluded that the surface roughness in the range of 0.4 to $7.2 \mu\text{m}$ (R_a) did not affect CHF unless a roughness factor, a ratio of liquid/solid contact area to the projected area under the liquid, was significantly larger than unity. However, various studies reported that CHF values increased considerably with increasing surface roughness for metallic surfaces, even up to 100% (Kim et al., 2016; Ramilison et al., 1992). Kim et al. (2016) prepared Cu surfaces with surface roughness range (R_a) of $0.041 \mu\text{m}$ to $2.36 \mu\text{m}$ by polishing with different sized grit papers. Pool boiling tests of the roughened Cu samples in water showed

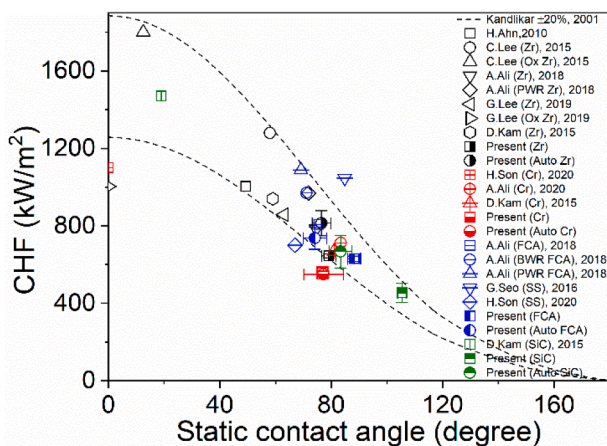


Fig. 8. CHF data for ATF candidate materials in pool boiling conditions as a function of static contact angle reported in literature and the present study. All experiments were performed with flat samples in the horizontal direction at atmospheric pressure. Predicted CHF plots based on the Kandlikar model with 20% uncertainty is also shown.

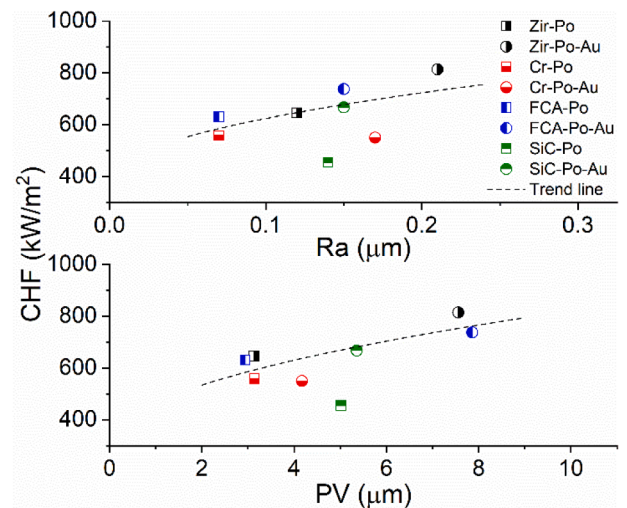


Fig. 9. CHF values as a function of roughness parameters of Zirlo®, Cr coated Zirlo®, FeCrAl coated Zirlo®, and CVD SiC flat heater materials with different surface conditions (as-polished and autoclave tested). The fitted lines based on a CHF correlation proposed by Kim et al. (Kim et al., 2016) are also shown in the plots.

augmented CHF. The authors believed CHF enhancement on rough surfaces is attributed to capillary wicking effect and proposed a CHF model by adding a capillary force term (which is a function of a surface roughness parameter) to the Kandlikar's model. The fitted lines in Fig. 9 are based on this correlation. Although additional experiments are needed to make a firm conclusion, the current data indicates that the surface roughness level of the ATF candidate materials is one factor influencing a CHF value in pool boiling experiments.

4.3. Surface contamination of heater surfaces

The very near-surface chemistry of the heater surfaces was evolved by not only the water autoclave corrosion test but also surface contamination during the pool boiling test and sample storage/handling. For example, carbon enrichment was identified on the autoclave tested samples as shown in the XPS results (Fig. 5). In particular, the high static contact angles for the autoclave tested Cr coatings and pool boiling tested CVD SiC may be attributed to the presence of a carbonaceous species on the surface including hydrophobic (i.e., low surface energies) hydrocarbon, graphite, and other organic compounds. Forrest et al., (Forrest et al., 2015) proposed that metallic surfaces with contact angle in the range of 50° to 85° are actually due to residual adsorbed carbon contaminants since superhydrophilicity ($\theta \sim 0^\circ$) is expected for completely clean metal and metal oxide surfaces due to the high surface energies. This is because the removal of the contaminants is generally very challenging with the typical solvent cleaning unless a plasma etching in vacuum chamber is utilized. In fact, it has been shown that static contact angle of stainless steel is a function of thickness of the organic contaminant (Mantel and Wightman, 1994). Metal oxides have their own specific rate and extent of organic contamination. Higher contamination rate on Cr-oxide surfaces was observed under dry and wet atmosphere compared to other oxides (Al-oxide, Si-oxide, Mo-oxide, Ni-oxide, Fe-oxide) (Matsunaga, 1979; Williams et al., 2008). Therefore, it is hypothesized that the greater extent of a thin carbon-rich overlayer on the smooth Cr coatings (with native Cr₂O₃ layer) was responsible for the high static contact angle and contributed to the slightly lower CHF values than the Zirlo® and FeCrAl samples. In CVD SiC samples, the formation of a carbon-rich layer is more complicated than the metallic samples due to chemical reactions at SiC/water interface during a boiling test or an autoclave test. It was reported that hydrothermal corrosion of SiC led to the formation of a thin film of carbon film due to the loss of Si at the surfaces (Gogotsi and Yoshimura, 1995; Henager et al., 2008; Terrani et al., 2015). It is believed that the chemical reaction occurred in the very near-surface regions of as-polished CVD SiC samples even during a boiling test, contributing to development of the carbon layer and eventually deteriorating the CHF.

4.4. Thermo-physical properties effect on CHF

There was no clear trend observed between the thermal activity and the measured CHF values in the present study. The CHF values for the Cr coated samples and the CVD SiC samples were lower than those of the bare Zirlo® samples although the coated samples and CVD SiC samples had approximately 27% and 600% higher thermal activity, respectively, compared to the bare Zirlo® samples. This result indicates that effects of surface chemistry (i.e., wettability) and surface roughness on CHF are more dominant than the effect of thermal activity.

5. Conclusions

The corrosion behavior and CHF of bare Zirlo®, Cr coatings, FeCrAl coatings, and bulk CVD SiC have been investigated. The coatings were deposited on flat Zirlo® substrates using cold spray technology. Some of the coated samples and the bulk materials were polished identically to maintain the surface roughness at similar levels. Autoclave tests for the as-prepared samples were performed at 360 °C water and 18.6 MPa for

15 days (360 h) to simulate corrosion of the ATF cladding materials in LWR normal operation. Both as-polished samples and post-autoclave samples were subjected to pool boiling experiments at atmospheric pressure to evaluate CHF values. The surface morphology, static contact angle, surface roughness, surface chemistry, and thermo-physical properties of the tested samples were measured to determine their effects on pool boiling CHF phenomenon. Surface morphology of Cr coated and FeCrAl coated samples were not changed after the autoclave tests while bare Zirlo® samples showed surface irregularities due to more extensive surface oxidation. The autoclave test altered surface wettability of the polished samples. The wettability change is attributed to evolution of near-surface chemistry due to oxidation and deposition of carbon-containing species. The extent of organic contamination including hydrocarbon and graphitic compounds varied depending on the material. CHF values of FeCrAl coatings were comparable to that of uncoated Zirlo® but Cr coatings exhibited a 13% lower CHF value. However, as-deposited Cr coatings showed 67% higher CHF than the as-polished Cr coatings due to its high surface roughness. Low CHF values were observed in as-polished CVD SiC samples compared to the metallic samples. CHF enhancement was generally observed for autoclave tested samples (bare Zirlo®, FeCrAl coatings, CVD SiC) and the samples showed almost similar CHF values. However, there was negligible change in CHF for Cr coatings after the autoclave test. These trends are explained by the evolution of near-surface chemistry, particularly the formation of deposition of a thin layer of hydrophobic carbonaceous compounds. Parametric analysis concluded that the effect of surface roughness and wettability on CHF are more dominant than thermo-physical properties of the heating material. The wettability can be changed by not only water corrosion but also surface contamination during boiling test and sample storage/handling. Based on the experimental results, it can be concluded that significant improvement of CHF in ATF cladding designs with the conventional surface finish is not expected unless surface engineering for the ATF cladding is performed to increase the wettability and the surface roughness. Prototypical pressurized flow boiling experiment with presence of radiation field is required for complete investigation of CHF phenomena for the various ATF designs.

CRedit authorship contribution statement

Hwasung Yeom: Conceptualization, Data curation, Writing - original draft. **Emilio Gutierrez:** Investigation, Data curation, Writing - review & editing. **HangJin Jo:** Funding acquisition, Methodology, Writing - review & editing. **Yimin Zhou:** Investigation. **Kaya Mondry:** Investigation. **Kumar Sridharan:** Funding acquisition, Writing - review & editing. **Michael Corradini:** Funding acquisition, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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